V2 / PICTURE WIRING

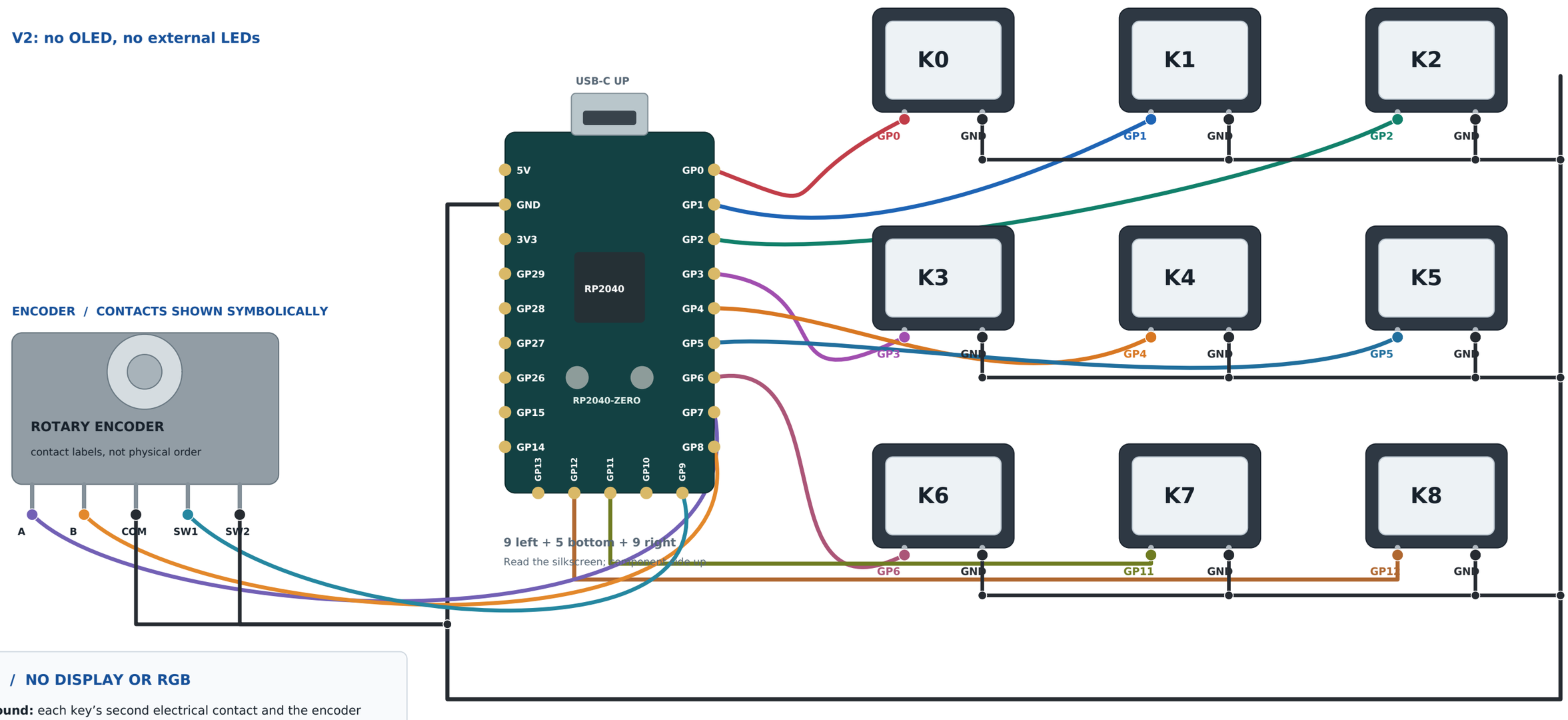
RP2040-Zero | 9 direct-wire switches + rotary encoder

Unplug power before soldering. Component drawings and wire routes are illustrative; identify the actual solder terminals with the part datasheet and a meter. Crossings are **not connected** unless a dot is shown. Read the actual board silkscreen.

NINE MECHANICAL SWITCHES / TOP LAYOUT

Drawn contacts are examples; use the two real electrical terminals.

V2: no OLED, no external LEDs



V2 / NO DISPLAY OR RGB

Ground: each key's second electrical contact and the encoder COM + SW2 share the black GND wire. No matrix diodes or extra power rail are needed.

Knob: A → GP7, B → GP8, push SW1 → GP9; COM and SW2 → GND. Identify the actual 5 contacts by datasheet or continuity.

Key layout: K0–K8 are shown left-to-right, top-to-bottom as viewed from above. The switch/encoder terminal positions are examples, **not a pinout**; verify them. Each key goes from its named GPIO to GND only when pressed. GP7/8/9 are encoder contacts, not key connections.

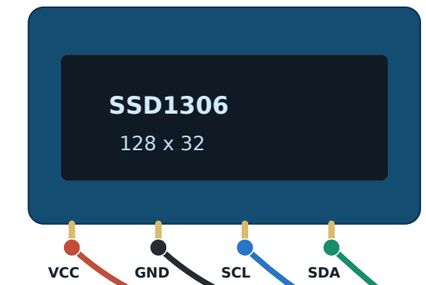
V1 / PICTURE WIRING

RP2040-Zero | 9 direct-wire switches + rotary encoder + OLED + planned RGB

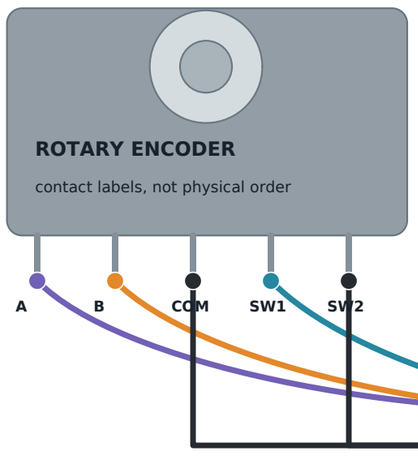
Unplug power before soldering. Component drawings and wire routes are illustrative; identify the actual solder terminals with the part datasheet and a meter. Crossings are **not connected** unless a dot is shown. Read the actual board silkscreen.

OLED: 3V3 / GND / GP28 SDA / GP29 SCL

3.3 V pull-ups; actual module pin order varies.

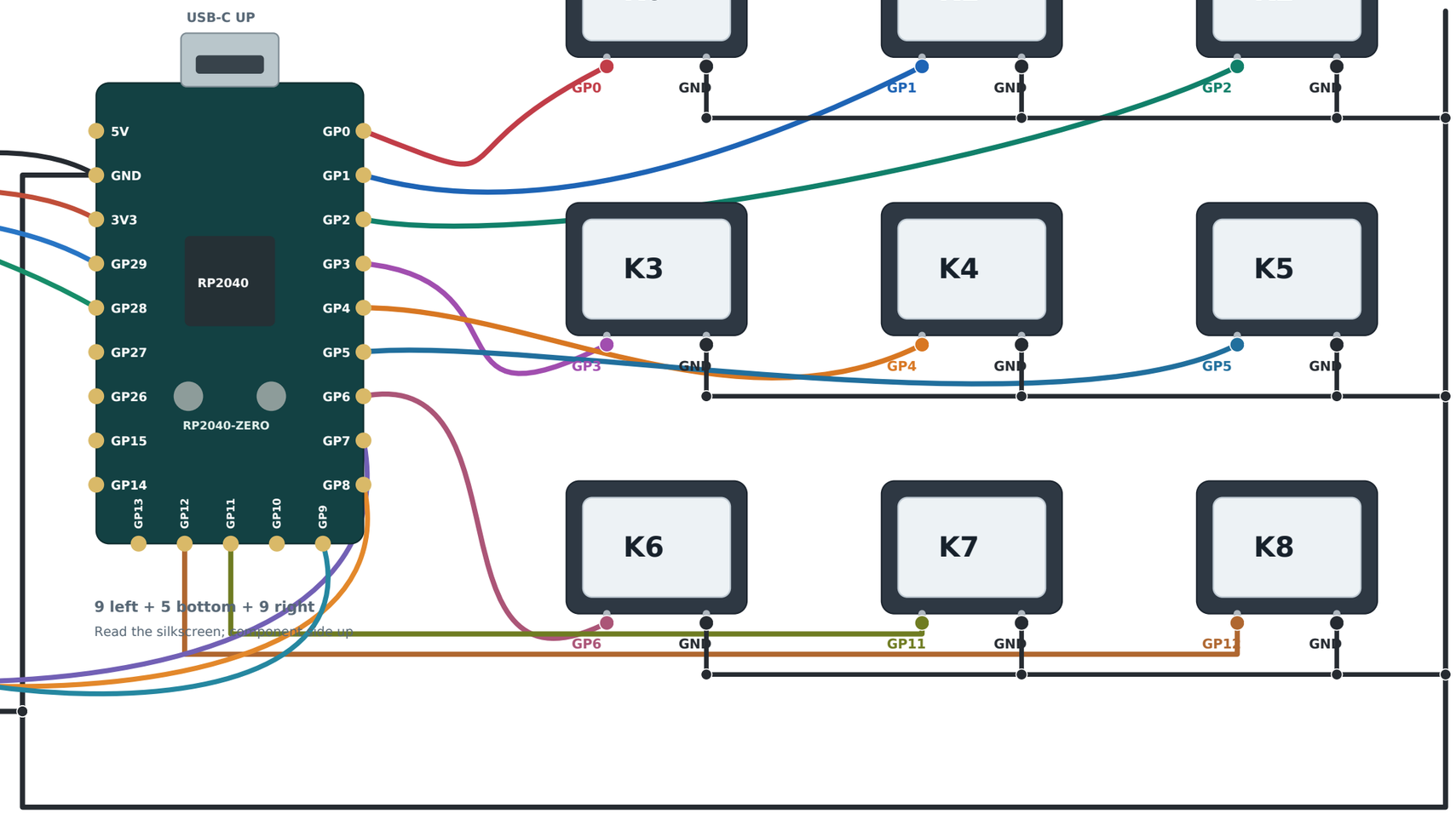


ENCODER / CONTACTS SHOWN SYMBOLICALLY



NINE MECHANICAL SWITCHES / TOP LAYOUT

Drawn contacts are examples; use the two real electrical terminals.



V1 RGB / NOT WIRED YET

Stop: LED strip, 5 V supply, current, isolation and firmware compatibility are not established. Do not connect LED power or GP10 yet. The separate guide shows the **proposed** 74AHCT1G125 + 330 Ω data path and required checks; it is not an approved power design.



Illustrative RGB strip: disconnected

Key layout: K0–K8 are shown left-to-right, top-to-bottom as viewed from above. The switch/encoder terminal positions are examples, **not a pinout**; verify them. Each key goes from its named GPIO to GND only when pressed. GP7/8/9 are encoder contacts, not key connections.